Graphical solutions to linear equation[redacted]inequalities

1 For each of the following equations:

- **i** Solve for the value of x that satisfies the equation.
- **ii** To verify the solution graphically, which two straight lines would need to be graphed?
- **iii** Graph the two lines in part (b) on the same plane.
- **iv** Hence find the value of x that satisfies the two linear equations simultaneously.

a $\dfrac{3x}{5} - 1 = 2$

b $2(x-1) - 3 = 7$

c $2x = 8 - 2x$

d $4x + 3 = 2x - 1$

e $2x + 3 = 6 - x$

f $2(2x+2) = -3(x+1)$

2 Using a graphing calculator, solve the following equations:

a $2x - 7 = -2x + 1$

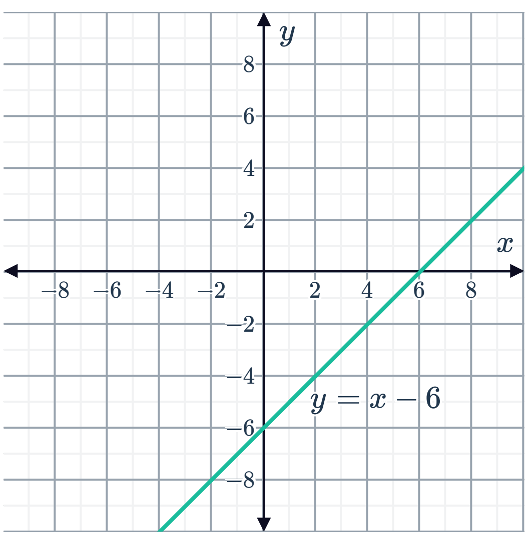
b $3x + 6 - x = 7x - 4$

c $8x - 3(7 - 2x) = 6(x - 4) + 5x + 9$

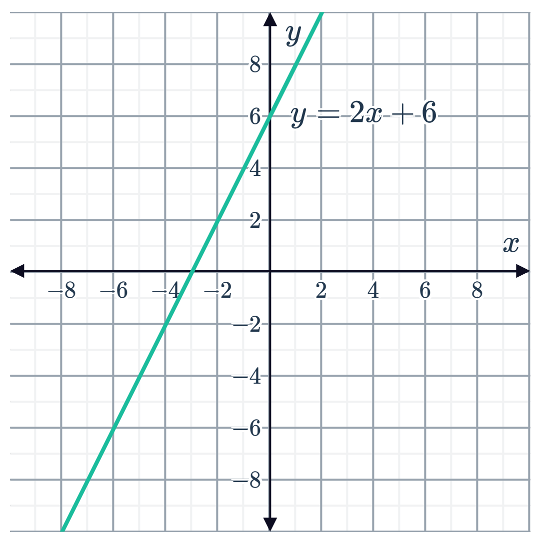
d $-4(3x + 8) = -7x + 8 - 5x$

3 Using the graphs, state the solution of the following inequalities:

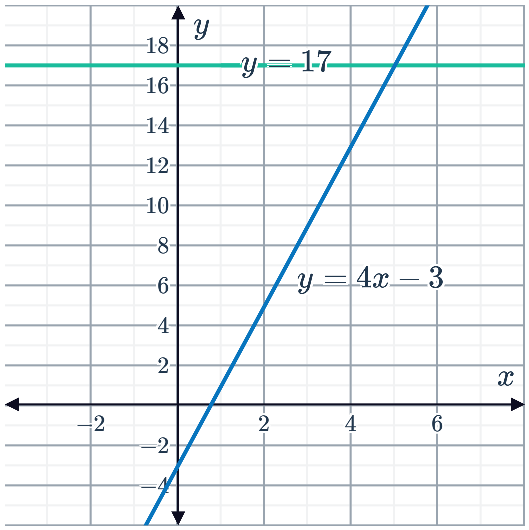
a $x - 6 < 0$



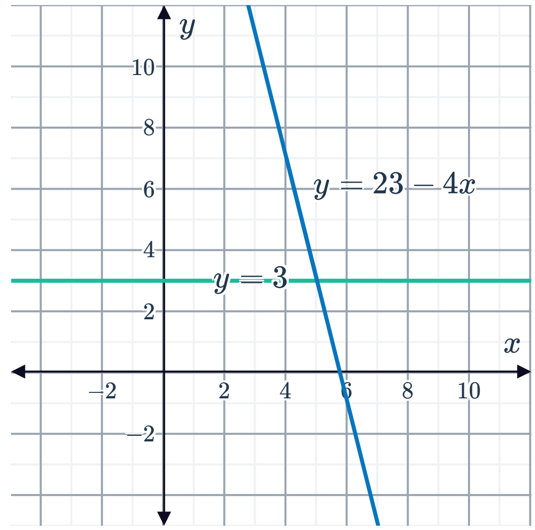
b $2x + 6 \geq 0$



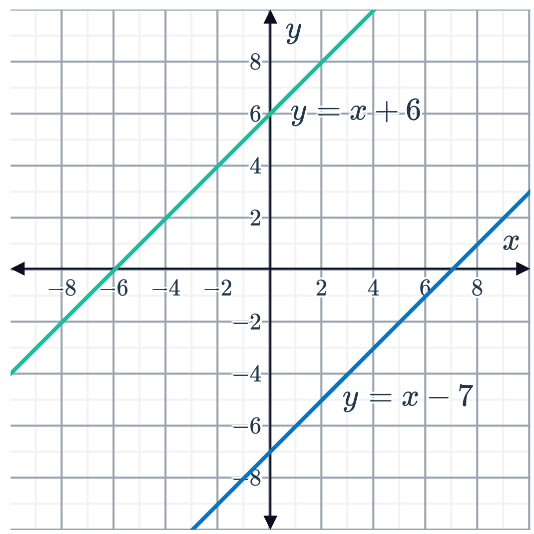
c $4x - 3 < 17$



$23 - 4x < 3$



4 Using the graphs of $y = x + 6$ and $y = x - 7$, find how many solutions the inequality $x + 6 \geq x - 7$ has.



5 To solve the inequality $x \leq 2x - 3$, Christa graphed $y = x + 3$. Find the other line she needs to graph to be able to solve the inequality graphically.

6 To solve the inequality $x \leq \dfrac{x-3}{4} - 1$, Tracy graphed $y = x - 3$. Find the other line she needs to graph to be able to solve the inequality graphically.

7 State whether the following inequalities can be solved graphically using the graphs of $y = 4x + 3$ and $y = 5 - x$:

a $(4x + 3) - (5 - x) \leq 0$

b $(4x + 3) + (3 - x) < 0$

c $\dfrac{4x + 3}{5} + x > 0$

d $3x + 8 < 0$

e $5x > 2$

8 Consider the linear function: $f\left(x\right) = \dfrac{-3}{2}x - 4$

a Write a linear function $g\left(x\right)$ such that $f\left(x\right) = g\left(x\right)$ has exactly one solution.

b With the $g\left(x\right)$ determined in part (a), find the exact solution of $f\left(x\right) = g\left(x\right)$.

9 Consider the linear equation $2y = -3x + 8$:

a Graph the linear equation by determining any two points on the line.

b This linear equation contains the point $(2, k)$. Find the value of k.